Does the reduced model's cost depend on bank width, and not on the mesh?

Yes for the mesh, partly for the bank width. Cost and accuracy are the same on every mesh we tried, in 2D and 3D, and that holds across retraining seeds and GPU types. Accuracy improves with bank width up to r = 384. But the method is not faster than the best conventional solver, a standard POD reduced model beats it, and it fails on steeper solutions than it was trained on. What the paper can stand on is mesh-independent cost, evaluation without a mesh, and memory.

E0 R_ST conditional Richardson estimates; development/validation cohorts only; sealed and OOD cohorts unopened. Total compute: 169 GPU-hours over 10 lanes.

The question the advisor asked

Cost does not depend on the mesh

The time to solve the reduced model stays flat as the simulation grid gets finer, in 2D and 3D.

`pass` 2D

8/8 settings pass; R = 1.01–1.02, largest 95 % upper end 1.04 (bar 1.25)

`pass` 3D

4/4 ROMs flat, max/min 1.01–1.09 (bar 1.25); quadrature check (Q2) of the four ROMs: C_old pass, C_W1024 fail, W_C_old fail, W_C_W768 fail — not all are qualified configurations

`pass (K = 16, incumbent)` 2D

K = 16: 1.06; incumbent K = 16: 1.03 (bar 1.25)

Accuracy does not depend on the mesh

For a fixed bank width, the error is the same on coarse and fine grids.

`pass` 2D

8/8 pass (every case within margin); R128 CN-R has a detectable small trend (Wilcoxon p = 0.00629): equivalence within margin, not zero dependence

`pass 21/30, inconclusive 9, no fail` 2D

inconclusive 4; inconclusive (unqualified configuration) 5; pass 21

Accuracy improves with bank width, up to a point

A wider bank helps up to r = 384; going to 512 does not help.

`pass` 2D

6/6 cells monotone

Cost against bank width

Cost grows with r, but not the way operation counts predict, so this registered prediction fails.

`fail` 2D component kernels at fixed (M

0/4 components follow the declared exponent; fitted p:

`fail` 2D

2/2 cells not monotone (384 → 512 does not improve)

`mixed` 2D

range 128–384: inconclusive 10, monotone 14, not monotone 3; range 128–512: inconclusive 7, not monotone 20

`fail` 2D

range not monotone

`inconclusive` 2D 1024²

too few numerically qualified seeds per K for a pooled verdict

N 1.17 (declared 1), J 1.08 (declared 1), JTJ 1.92 (declared 2), chol 1.24 (declared 3)

`fail` 3D component kernels at fixed (M

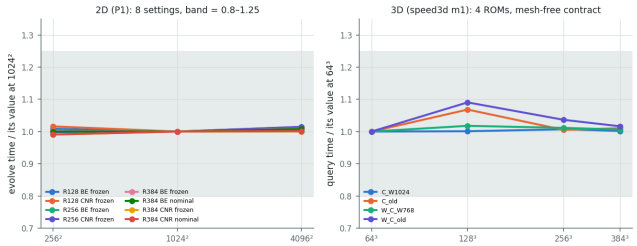
1/4 components follow the declared exponent; fitted p: N 1.01 (declared 1), J 1.03 (declared 1), JTJ 1.33 (declared 2), chol 1.03 (declared 3)

`descriptive` 2D full query

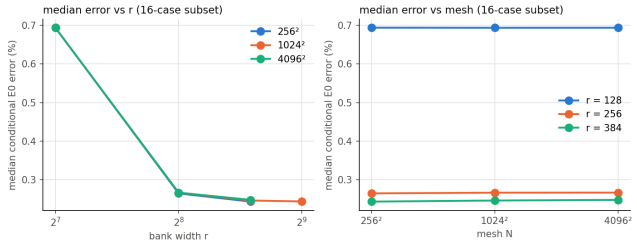
elasticity of query time in r 0.654, in M 0.528, in m 0.653

`inconclusive` 2D

exponent identified at 1 of 3 meshes (4096²: p = 0.119, seed 0 only)



Solve time against mesh size: flat lines mean the cost does not depend on the mesh



Error against bank width, one line per mesh: the lines lie on top of each other

Against other methods

Against the best full solver (K2)

A spectral solver on a coarse grid reaches the same accuracy faster, in 2D and 3D. The plan's kill criterion fires.

`fires` 2D 4096²

fires under the registered dev6 workload rule (A3.1); union speed-up 0.0493–0.706 over 21 configurations (largest interval end 0.74)

`fires` 3D L = 384

union speed-up 0.00786–0.158 (4 ROMs)

`fail` 2D 1024²

ROM < 2x ×21

Against a standard reduced model (H-base)

A tuned POD model with hyper-reduction is faster and has lower worst-case error for our 2D settings.

`fail` 2D 256²

8/8 of our configurations dominated (faster and lower worst error) by a registered POD pick; restrictions: saturation check failed (A1.3); budget: budget band unresolved; developmental, in-sample, unadjusted

`fail` 2D 1024²

8/8 of our configurations dominated (faster and lower worst error) by a registered POD pick; restrictions:

`fail` 2D 4096²

ROM < 2x ×21

`fail` 3D L = 384

ROM < 2x ×4

snapshot-limited check failed (A1.3); budget: budget band unresolved; developmental, in-sample, unadjusted

`fail where assessed` 3D 65³

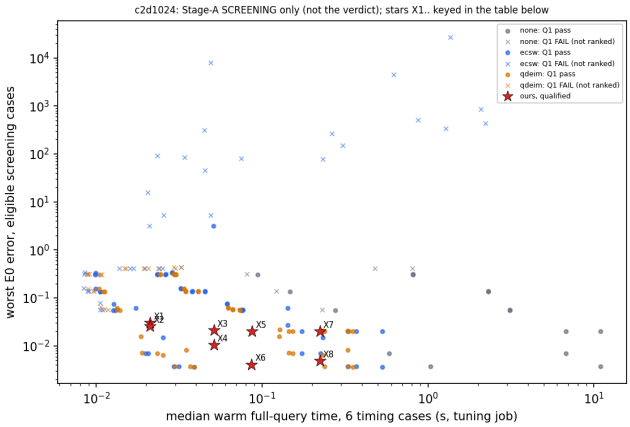
2/4 of our configurations dominated (faster and lower worst error) by a registered POD pick; 2 neutral conclusions withheld; restrictions: basis adequacy failed (A2.2); ECSW coverage incomplete (A3.11); budget: unequal-budget, descriptive; developmental, in-sample, unadjusted

`fail` 3D 65³

5/5 of our configurations dominated (faster and lower worst error) by a registered POD pick; restrictions: basis adequacy failed (A2.2); ECSW coverage incomplete (A3.11); budget: unequal-budget, descriptive; developmental, in-sample, unadjusted



Speed-up over the best full solver at each mesh (below 1 means the full solver is faster)



2D 1024²: accuracy against cost for POD baselines and our model

How robust it is

New regimes (outside the training range)

On steeper fronts than in training, the model is not qualified; the bank itself cannot represent them.

`HOLDS ×1, IMPROVES ×1, INCONCLUSIVE ×6` 2D 1024²

Random quadrature points

Random choices of quadrature points change little; the 2D 'fragile' cells are about rule convergence, not accuracy.

`H-E1a fragile ×6, inconclusive ×6` 2D

H-E1b typical ×12; "fragile" = successor-rule

qualified 8/8; labels HOLDS ×1, IMPROVES ×1, INCONCLUSIVE ×6; median error 0.246 %–1.37 %; ratio to in-range baseline 0.55–1.53

`DEGRADES ×2, INCONCLUSIVE ×6` 2D 1024²

qualified 8/8; labels DEGRADES ×2, INCONCLUSIVE ×6; median error 0.883 %–4.17 %; ratio to in-range baseline 1.63–9.25

`NOT QUALIFIED ×8` 2D 1024²

qualified 0/8; labels NOT QUALIFIED ×8; median error 5.2 %–11.2 %; ratio to in-range baseline 5.78–36.7

`NOT QUALIFIED ×8` 2D 1024²

qualified 0/8; labels NOT QUALIFIED ×8; median error 12.7 %–25.1 %; ratio to in-range baseline 14.2–131

`HOLDS ×3, INCONCLUSIVE ×5` 2D 1024²

qualified 8/8; labels HOLDS ×3, INCONCLUSIVE ×5; median error 0.2 %–1.15 %; ratio to in-range baseline 0.989–1.55

`INCONCLUSIVE ×8` 2D 1024²

qualified 8/8; labels INCONCLUSIVE ×8; median error 0.245 %–1.4 %; ratio to in-range baseline 1.11–1.64

`DEGRADES ×8 (descriptive)` 2D 1024²

qualified 8/8; labels DEGRADES ×8; median error 1.22 %–4.22 %; ratio to in-range baseline 2.97–9.79

convergence only, see section 3.11

`H-E1a robust ×2` 3D

H-E1b atypically unfavourable ×1, typical ×1

Retraining with new seeds

Retraining with new seeds keeps mesh-independence and the bank-width trend, but the best quadrature rule changes per bank.

`seed-robust 140/183` 2D

all rows: inconclusive ×22, seed-dependent ×21, seed-robust ×140; headline accuracy rows: seed-robust ×24; selected-rule rows: seed-dependent ×3

`seed-robust 45/129` 3D

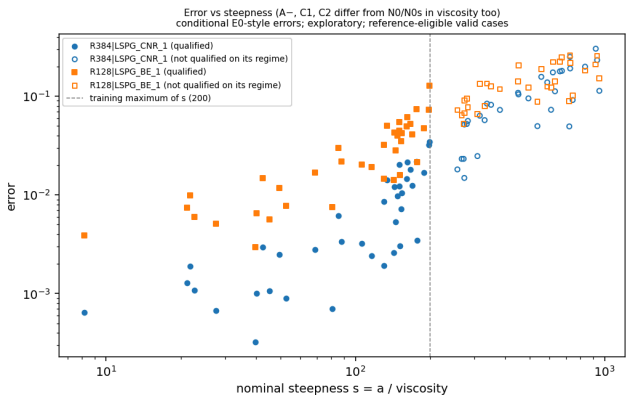
all rows: inconclusive ×67, seed-dependent ×17, seed-robust ×45; headline accuracy rows: inconclusive ×22, seed-robust ×10; selected-rule rows: inconclusive ×12, seed-dependent ×2, seed-robust ×10

Different GPUs

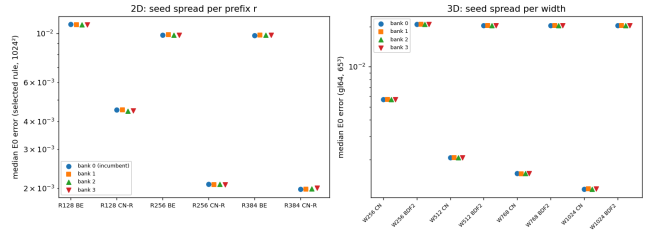
A100 and H200 GPUs give the same verdicts.

`inconclusive ×1, verdict-robust ×5` A100-80G vs H200

φ intervals: hardware-consistent ×6, hardware-dependent ×4



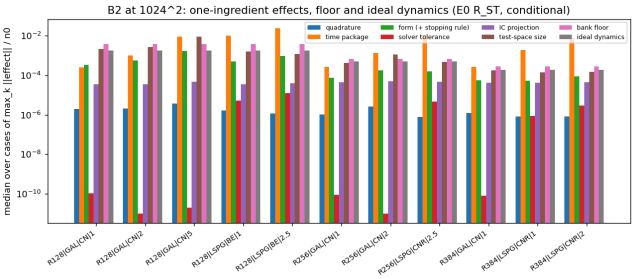
Error grows with front steepness beyond the training range



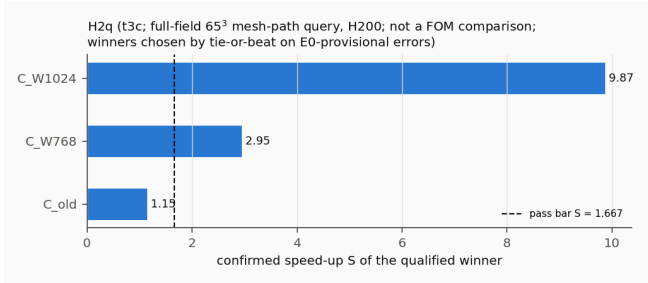
Spread of results across retraining seeds

Where the error comes from

The time-stepping error dominates for the standard schemes. Results stay stable when runs are extended in time, and the scoring grid is fine enough.



Error budget at 1024²: which ingredient dominates



3D: speed-up of the cheapest fully checked setting over our earlier setups (reduced model against reduced model)

Every verdict

▼ Show all 48 rows

Hypothesis	Scope	Verdict	Key number
H-cost-N	2D, 256²–4096², R128–R384, BE/CN-R, frozen + nominal mode lists; evolution time with internal-state recording (not the full query)	pass	8/8 settings pass; R = 1.01–1.02, largest 95 % upper end 1.04 (bar 1.25)
H-cost-N	3D, 64³–384³ (H4.1), mesh-free contract, deployment mode policy, 4 ROMs	pass	4/4 ROMs flat, max/min 1.01–1.09 (bar 1.25); quadrature check (Q2) of the four ROMs: C_old pass, C_W1024 fail, W_C_old fail,

			W_C_W768 fail — not all are qualified configurations
H-cost-N	2D, 256²–4096², seed 0 heads, BE, evolution only	`pass (K = 16, incumbent); not reached for K = 4, 8, 32`	K = 16: 1.06; incumbent K = 16: 1.03 (bar 1.25)
H-cost-r	2D component kernels at fixed (M, m)	`fail`	0/4 components follow the declared exponent; fitted p: N 1.17 (declared 1), J 1.08 (declared 1), JTJ 1.92 (declared 2), chol 1.24 (declared 3)
H-cost-r	3D component kernels at fixed (M, m)	`fail`	1/4 components follow the declared exponent; fitted p: N 1.01 (declared 1), J 1.03 (declared 1), JTJ 1.33 (declared 2), chol 1.03 (declared 3)
H-cost-r	2D full query, t2 timing panel (descriptive elasticities, no verdict)	`descriptive`	elasticity of query time in r 0.654, in M 0.528, in m 0.653
H-cost-r	2D, cost against K = 4–32 (descriptive)	`inconclusive`	exponent identified at 1 of 3 meshes (4096²: p = 0.119, seed 0 only)
H-acc-N	2D, 256² vs 1024² (38 cases) and 4096² vs 1024² (16 cases), P1 settings	`pass`	8/8 pass (every case within margin); R128 CN-R has a detectable small trend (Wilcoxon p = 0.00629): equivalence within margin, not zero dependence
H-acc-N	2D, extended scheme / step set (B1, 30 tests)	`pass 21/30, inconclusive 9, no fail`	inconclusive 4; inconclusive (unqualified configuration) 5; pass 21
H-acc-r	2D, r = 128 → 256 → 384, P1 cells (256², 1024², 4096² × BE, CN-R)	`pass`	6/6 cells monotone
H-acc-r	2D, registered range r = 128–512 (1024² only)	`fail`	2/2 cells not monotone (384 → 512 does not improve)
H-acc-r	2D, extended scheme / step set (B1, 27 cells)	`mixed`	range 128–384: inconclusive 10, monotone 14, not monotone 3; range 128–512: inconclusive 7, not monotone 20
H-acc-r	2D, fixed (M, m) = (2048, Gauss 256²) (A3)	`fail`	range not monotone
H-acc-r	2D 1024², K = 4 → 8 → 16 → 32 (BE)	`inconclusive`	too few numerically qualified seeds per K for a pooled verdict

H-comp	2D 1024², vs upwind + BE, 21 qualified LSPG configurations	`mixed (pass only against the screening-selected comparator)`	ROM < 2x ×2, ROM >= 2x ×16, unresolved ×3
H-comp	2D 1024², vs central + CN-R, 21 qualified LSPG configurations	`fail or unresolved`	ROM < 2x ×19, unresolved ×2
H-comp	2D 1024², vs sine pseudo-spectral, 21 qualified LSPG configurations	`fail`	ROM < 2x ×21
H-comp	2D 1024², vs union of families, 21 qualified LSPG configurations	`fail`	ROM < 2x ×21
H-comp	2D 4096², vs upwind + BE, 21 qualified LSPG configurations	`mixed (pass only against the screening-selected comparator)`	ROM < 2x ×2, ROM >= 2x ×16, unresolved ×3
H-comp	2D 4096², vs central + CN-R, 21 qualified LSPG configurations	`fail or unresolved`	ROM < 2x ×19, unresolved ×2
H-comp	2D 4096², vs sine pseudo-spectral, 21 qualified LSPG configurations	`fail`	ROM < 2x ×21
H-comp	2D 4096², vs union of families, 21 qualified LSPG configurations	`fail`	ROM < 2x ×21
K2	2D 4096², registered LSPG schemes vs best matched FOM (union)	`fires`	fires under the registered dev6 workload rule (A3.1); union speed-up 0.0493–0.706 over 21 configurations (largest interval end 0.74)
H-comp	3D L = 384, vs upwind + BE, 4 selected ROMs (not all quadrature-qualified)	`inconclusive (no ladder member as accurate)`	no ladder match ×4
H-comp	3D L = 384, vs upwind + CN-R, 4 selected ROMs (not all quadrature-qualified)	`inconclusive (no ladder member as accurate)`	no ladder match ×4
H-comp	3D L = 384, vs central + CN-R, 4 selected ROMs (not all quadrature-qualified)	`mixed`	ROM < 2x ×3, ROM >= 2x ×1
H-comp	3D L = 384, vs sine pseudo-spectral, 4 selected ROMs (not all quadrature-qualified)	`fail`	ROM < 2x ×4
H-comp	3D L = 384, vs union of families, 4 selected ROMs (not all	`fail`	ROM < 2x ×4

	quadrature-qualified)		
K2	3D L = 384, selected ROMs vs best matched FOM (union)	`fires`	union speed-up 0.00786–0.158 (4 ROMs)
H-base	2D 256^2	`fail`	8/8 of our configurations dominated (faster and lower worst error) by a registered POD pick; restrictions: saturation check failed (A1.3); budget: budget band unresolved; developmental, in-sample, unadjusted
H-base	2D 1024^2	`fail`	8/8 of our configurations dominated (faster and lower worst error) by a registered POD pick; restrictions: snapshot-limited check failed (A1.3); budget: budget band unresolved; developmental, in-sample, unadjusted
H-base	3D 65^3, qualified subset (batch 2)	`fail where assessed; neutral conclusions withheld`	2/4 of our configurations dominated (faster and lower worst error) by a registered POD pick; 2 neutral conclusions withheld; restrictions: basis adequacy failed (A2.2); ECSW coverage incomplete (A3.11); budget: unequal-budget, descriptive; developmental, in-sample, unadjusted
H-base	3D 65^3, diagnostic (batch 1)	`fail`	5/5 of our configurations dominated (faster and lower worst error) by a registered POD pick; restrictions: basis adequacy failed (A2.2); ECSW coverage incomplete (A3.11); budget: unequal-budget, descriptive; developmental, in-sample, unadjusted
H2q	3D, qualified winner vs our earlier configuration C_old (ROM vs ROM; bar 1/0.6); full-field 65^{3} mesh-path query, H200; not a comparison with the best FOM (K2 already fired); qualified vs qualified comparator	`fail (the selected candidate failed confirmation)`	S = 1.15 (comparator Q2: pass); the unqualified speed3d winners had 2.01
H2q	3D, qualified winner vs our earlier configuration C_W768 (ROM vs ROM; bar 1/0.6); full-field	`pass`	S = 2.95 (comparator Q2: pass); the unqualified speed3d winners had 5.37

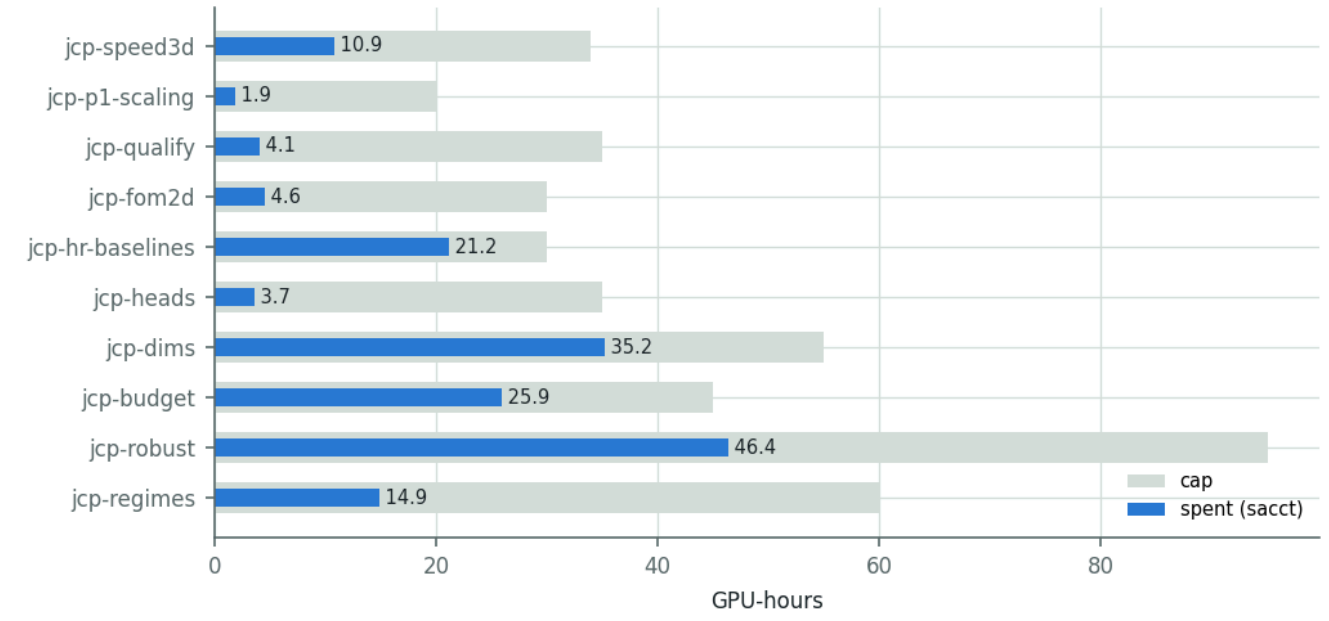
	65^{3} mesh-path query, H200; not a comparison with the best FOM (K2 already fired); qualified vs qualified comparator		
H2q	3D, qualified winner vs our earlier configuration C_W1024 (ROM vs ROM; bar 1/0.6); full-field 65^{3} mesh-path query, H200; not a comparison with the best FOM (K2 already fired); qualified configuration vs the deployed (unqualified) comparator	`pass`	S = 9.87 (comparator Q2: fail); the unqualified speed3d winners had 17.2
E1 rule randomisation	2D, 12 registered cells, 16 shifted draws each	`H-E1a fragile ×6, inconclusive ×6; H-E1c insensitive ×12`	H-E1b typical ×12; "fragile" = successor-rule convergence only, see section 3.11
E1 rule randomisation	3D, 2 registered cells, 16 shifted draws each	`H-E1a robust ×2; H-E1c insensitive ×2`	H-E1b atypically unfavourable ×1, typical ×1
E2 recipe seeds	2D, seeds 1–3, 183 verdict rows	`seed-robust 140/183`	all rows: inconclusive ×22, seed-dependent ×21, seed-robust ×140; headline accuracy rows: seed-robust ×24; selected-rule rows: seed-dependent ×3
E2 recipe seeds	3D, seeds 1–3, 129 verdict rows	`seed-robust 45/129`	all rows: inconclusive ×67, seed-dependent ×17, seed-robust ×45; headline accuracy rows: inconclusive ×22, seed-robust ×10; selected-rule rows: inconclusive ×12, seed-dependent ×2, seed-robust ×10
E5 hardware	A100-80G vs H200, 4 jobs on 4 physical GPUs; frozen H200-selected FOM comparators; registered timing verdicts	`inconclusive ×1, verdict-robust ×5`	φ intervals: hardware-consistent ×6, hardware-dependent ×4
D1 regime	2D 1024^2, A+ (v above the training range), 8 primary LSPG configurations; viscosity extrapolation confounded with steepness	`HOLDS ×1, IMPROVES ×1, INCONCLUSIVE ×6`	qualified 8/8; labels HOLDS ×1, IMPROVES ×1, INCONCLUSIVE ×6; median error 0.246 %–1.37 %; ratio to in-range baseline 0.55–1.53
D1 regime	2D 1024^2, A– (v below the training range), 8 primary LSPG configurations	`DEGRADES ×2, INCONCLUSIVE ×6`	qualified 8/8; labels DEGRADES ×2, INCONCLUSIVE ×6; median error 0.883 %–4.17 %; ratio to in-range baseline 1.63–9.25

D1 regime	2D 1024², C1 (low v, high steepness s; paired with A−), 8 primary LSPG configurations	`NOT QUALIFIED ×8`	qualified 0/8; labels NOT QUALIFIED ×8; median error 5.2 %–11.2 %; ratio to in-range baseline 5.78–36.7
D1 regime	2D 1024², C2 (lower v, higher s), 8 primary LSPG configurations	`NOT QUALIFIED ×8`	qualified 0/8; labels NOT QUALIFIED ×8; median error 12.7 %–25.1 %; ratio to in-range baseline 14.2–131
D1 regime	2D 1024², B1 (IC family: two same-sign bumps), 8 primary LSPG configurations	`HOLDS ×3, INCONCLUSIVE ×5`	qualified 8/8; labels HOLDS ×3, INCONCLUSIVE ×5; median error 0.2 %–1.15 %; ratio to in-range baseline 0.989–1.55
D1 regime	2D 1024², B2 (IC family: signed pair), 8 primary LSPG configurations	`INCONCLUSIVE ×8`	qualified 8/8; labels INCONCLUSIVE ×8; median error 0.245 %–1.4 %; ratio to in-range baseline 1.11–1.64
D1 regime	2D 1024², N0s (stiff corner of the training box; descriptive), 8 primary LSPG configurations	`DEGRADES ×8 (descriptive)`	qualified 8/8; labels DEGRADES ×8; median error 1.22 %–4.22 %; ratio to in-range baseline 2.97–9.79

Corrections made during the campaign

- The first 3D speed-ups over our own earlier setups used settings that failed the quadrature check. Re-timed with fully checked settings, a real gain remains against the W768 setup, but against the original deployment it is below the bar.
- An earlier large 2D speed-up held only against a same-mesh upwind solver, not against the best full solver.
- Some earlier "flat accuracy across meshes" comparisons used slightly different test functions on the 256² mesh. Fixing them moved errors by hundredths of a percentage point.
- Several lanes' first analyses had errors (wrong dominance rule, under-counted budgets, double-counted GPU-hours). Each was caught by an audit, corrected and re-audited. The full list is in the report.

Compute and lanes



GPU-hours per lane

▼ GPU-hours per lane

Lane	Spent	Cap
jcp-speed3d	10.9	34
jcp-p1-scaling	1.9	20
jcp-qualify	4.1	35
jcp-fom2d	4.6	30
jcp-hr-baselines	21.2	30
jcp-heads	3.7	35
jcp-dims	35.2	55
jcp-budget	25.9	45
jcp-robust	46.4	95
jcp-regimes	14.9	60

▼ Branches awaiting your merge decision

Lane	Branch	Last commit	Mirror
jcp-speed3d	`exp/2026-10-14-jcp-speed3d`	`691465d88`	`6c270e6c5`
jcp-p1-scaling	`exp/2026-10-17-jcp-p1-scaling`	`5b782026b`	`3b589dad5`
jcp-qualify	`exp/2026-10-18-jcp-qualify`	`250b20d91`	`35620c70e`
jcp-fom2d	`exp/2026-10-18-jcp-fom2d`	`12103d331`	`c4089abe6`
jcp-hr-baselines	`exp/2026-10-06-jcp-hr-baselines`	`2e3abac47`	`2fa5b5967`

jcp-heads	`exp/2026-10-18-jcp-heads`	`0ba9a1288`	`ed3d0c841`
jcp-dims	`exp/2026-10-18-jcp-dims`	`a99906fe6`	`038cce675`
jcp-budget	`exp/2026-10-18-jcp-budget`	`d70608191`	`9be95f4fb`
jcp-robust	`exp/2026-10-19-jcp-robust`	`cbf574373`	`d49bd0085`
jcp-regimes	`exp/2026-10-19-jcp-regimes`	`70ce5f0f2`	`ee52292fe`

Words used here

- **Bank width r**: how many basis coefficients the reduced model solves for. This is the "k" in the advisor's question; the other reading, the nonlinear head's size K, is also tested.
- **Mesh**: grid points per side of the full simulation.
- **Full solver / spectral solver**: the conventional simulation; "spectral" uses fast Fourier-type transforms.
- **K2**: the plan's pre-registered outcome "we are not at least 2× faster than the best full solver".
- **POD baseline**: the standard reduced model built from simulation snapshots, with hyper-reduction (ECSW or Q-DEIM) to make it fast.
- **Qualified**: passed the solver and quadrature checks, so its numbers can be trusted.
- **Seed**: the random starting point of training; a seed-robust result holds for every retrained model.
- **Development / validation cases**: the test problems used for choosing and checking; the sealed final test set is still unopened.